Error-Corrected Heterogeneous Treatment Effects: When LLM Annotation Debiasing Hurts

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Abstract

We prove that correcting LLM annotation error with a single pooled confusion matrix—a common approach when LLM labels serve as effect modifiers in heterogeneous treatment effect (HTE) analysis—introduces subgroup-level bias proportional to $\delta_s \cdot \Delta\tau$, the product of each subgroup's misclassification deviation ($\delta_s = \pi_s - \bar{\pi}$) and CATE heterogeneity, an exact identity verified at $R^2 \geq 0.97$ on both synthetic grids and 11 real LLM confusion matrices. In 50% of K=2 configurations (260/520) across our parameter grid, global correction worsens subgroup CATE bias, with amplification up to 2.86× (difference-in-means) and 5.9× (K=4, CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) addresses this through hierarchical Bayesian subgroup-stratified confusion matrix estimation. Under extreme misclassification heterogeneity (K=2), EC-HTE reduces |bias| from 0.059 to 0.010 with nominal coverage (≥0.91 across all configurations). Independently, the bias identity yields a computable diagnostic (AUC = 0.86) that flags when global correction increases bias using only per-subgroup misclassification rates—actionable without adopting EC-HTE. Composed with Prediction-Powered Conditional Inference (PPCI) for dual covariate/outcome error, the two corrections are each necessary and jointly sufficient, reducing |bias| from 0.179 to 0.007. EC-HTE's advantage is conditional on sufficient error heterogeneity and per-subgroup expert budget; in the semi-synthetic TweetEval experiment (moderate $\Delta\pi$, unbalanced subgroups), global correction matches EC-HTE on bias with lower aggregate RMSE (0.124 vs. 0.225). We validate across 520 synthetic configurations (100 Monte Carlo runs each), a semi-synthetic TweetEval experiment (N=5,000), and eight LLMs (open-source 7B–9B and proprietary) across two NLP datasets, confirming both $\Delta\pi$ heterogeneity and the bias identity ($R^2 = 0.975$) on real annotations.

1 Introduction

Large language models (LLMs) now routinely replace human annotators in computational social science (CSS), producing labels for sentiment, hate speech, stance, and political discourse at costs orders of magnitude below manual coding (Gilardi et al., 2023; Törnberg, 2025; Ziems et al., 2024). Yet these annotations are not uniformly accurate. Ashwin et al. (2025) document that LLM misclassification rates vary by up to 5× across demographic subgroups defined by refugee status, gender, and education level. In a pilot study, GPT-4.1 achieves 85.8% binary accuracy on TweetEval sentiment, but error rates range from 10.2% to 20.0% across text-feature subgroups ($\Delta\pi = 0.098$). When such annotations serve as effect modifiers in heterogeneous treatment effect (HTE) analysis, subgroup-specific errors propagate directly into conditional average treatment effect (CATE) estimates. The resulting bias is not random noise that averages out; it is structured by the same demographic axes that define the research question. Standard correction procedures[1], which estimate a single confusion matrix and apply its inverse uniformly, cannot account for this heterogeneity.

We formalize this problem and derive an exact bias identity for the correction-induced error. When a pooled confusion matrix is applied to correct subgroup CATEs, it misattributes error across subgroups with an exact product form: the bias in subgroup s equals $\delta_s \cdot \Delta\tau/(1 - 2\bar{\pi})$, where δ_s is the deviation from the pooled error rate and $\Delta\tau$ is the CATE heterogeneity (Theorem 1). This is not a bound but an exact population-level identity, verified empirically with $R^2 = 0.994$ (K=2) and $R^2 = 0.974$ (K=4) on synthetic grids, and $R^2 = 0.975$ across 11 real LLM confusion matrices spanning open-source and proprietary model

[1]Throughout, "debiasing" refers to correction of annotation measurement-error bias, not algorithmic fairness debiasing.

families. A direct consequence is that global correction can be worse than no correction at all: in 50% of $K{=}2$ configurations (260/520) across our parameter grid, global correction increases absolute bias relative to the naive estimator, with amplification reaching 2.86× ($K{=}2$) and 5.9× ($K{=}4$ with CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) addresses this through hierarchical Bayesian subgroup-stratified confusion matrix estimation, reducing |bias| from 0.059 to 0.010 under extreme heterogeneity ($K{=}2$). The bias decomposition identity also yields a $\Delta\tau$-free diagnostic (Proposition 3): a condition on per-subgroup misclassification rates alone predicts when global correction will increase bias (AUC = 0.86)—independently useful even without adopting EC-HTE, since practitioners can flag harmful correction from a pilot sample before estimating any treatment effects. EC-HTE's advantage is conditional on sufficient error heterogeneity and expert budget; when $\Delta\pi$ is small, the variance cost of stratification outweighs its bias benefit—the expected bias-variance tradeoff of any subgroup-stratified correction (Figure 1).

Subgroup CATEs inform resource allocation: whether a content moderation policy reduces toxicity more in long-form versus short-form posts, or whether a public health campaign reaches different demographic groups equally. Biased CATE estimates can reverse the sign of apparent treatment heterogeneity, leading practitioners to allocate resources away from communities that benefit most. The failure mode is subtle: a researcher who corrects for annotation noise using a pooled confusion matrix may introduce more bias than doing nothing, precisely in the low-error subgroups where original labels were most trustworthy.

Existing corrections—PPI (Angelopoulos et al., 2023), DSL (Egami et al., 2024), PPCI (Sui and Candès, 2026), arm-specific PPI (Mozer, 2026), ForestIV (Bu et al., 2023)—address population-level or outcome-side measurement error but assume homogeneous prediction error across subgroups; none corrects bias when the mismeasured variable is the effect modifier indexing the CATE function (Section 2).

Our contributions are:

- A bias decomposition identity (Theorem 1) proving that global correction introduces subgroup bias proportional to $\delta_s \cdot \Delta\tau$ (per-subgroup misclassification deviation × CATE heterogeneity), with a $\Delta\tau$-free diagnostic (Proposition 3; AUC = 0.86) that flags when correction is harmful using only per-subgroup misclassification rates—independently actionable without EC-HTE (Sections 3, 5.4).
- EC-HTE, the natural correction prescribed by the bias structure: per-subgroup confusion matrix estimation via hierarchical Bayes, compatible with any CATE estimator and composable with PPCI for dual covariate/outcome error (|bias|: 0.179 → 0.007; Sections 4–5.5).
- Applicability conditions and crossover analysis ($n_{\text{expert}}{=}250$ for $K{=}2$, $n_{\text{expert}}{=}1{,}000$ for $K{=}4$) establishing when EC-HTE improves over global correction and when it does not (Appendix I).

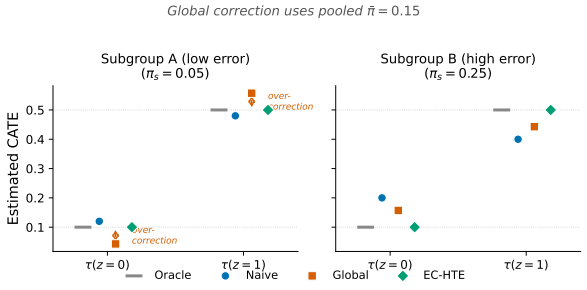


Figure 1: Bias mechanism under global vs. subgroup-stratified correction. Global correction applies the pooled error rate $\bar{\pi}$ uniformly, over-correcting low-error subgroups (introducing bias where none existed) and under-correcting high-error subgroups. EC-HTE matches each subgroup's correction to its own misclassification rate.

2 Related Work

Measurement error and annotation models. Classical measurement error theory shows that correction via estimated error models can amplify MSE when the error structure is misspecified (Fuller, 1987; Carroll et al., 2006); Bayesian approaches require informative priors when identification is weak (Gustafson, 2003), and nonparametric methods use repeated measurements or instruments to recover the true distribution (Hu and Schennach, 2008).

Three distinctions separate our contribution from prior measurement error analyses. First, we study effect modifier misclassification—the mismeasured variable defines the CATE function itself—rather than regressor (Frazis and Loewenstein, 2003; Carroll et al., 2006) or confounder (Ogburn and Vander-

Weele, 2012; Greenland, 1980) error, where attenuation is the primary concern. Second, Theorem 1 derives an exact identity for correction-induced bias—bias created by the debiasing procedure—rather than bounding the attenuation from misclassification itself; global correction introduces structured bias absent from the uncorrected estimator. Third, the $\delta_s \cdot \Delta\tau$ product structure arises only under heterogeneous error rates across subgroups; all homogeneous-error analyses (Battistin and Chesher, 2014; Fong and Tyler, 2021) miss this interaction entirely. In the annotation literature, Dawid and Skene (1979) introduced confusion matrix estimation for multiple annotators, with extensions to item difficulty, annotator competence, and hierarchical priors (Passonneau and Carpenter, 2014; Hovy et al., 2013; Paun et al., 2018). Our hierarchical estimator (Section 4) adapts this architecture from label aggregation to causal inference, tying the prior structure to the subgroup dimension that defines treatment effect heterogeneity.

HTE estimation and LLM annotation error. Causal forests (Athey and Imbens, 2016; Wager and Athey, 2018), double/debiased ML (Chernozhukov et al., 2018), doubly-robust CATE estimators (Kennedy, 2023), and meta-learners (Künzel et al., 2019) all assume that covariates defining heterogeneity are observed without error; when the effect modifier is misclassified, CATE estimates conflate effects across groups. EC-HTE provides upstream correction that restores valid inputs to any downstream estimator. The corrected pseudo-outcomes are in principle compatible with other HTE estimators such as the R-learner (Nie and Wager, 2021). For broader perspectives on causal inference with text data, see Feder et al. (2022) and Keith et al. (2020); Wang et al. (2020) study inference correction when outcomes are ML-predicted, a related but distinct problem from effect modifier misclassification. LLMs now match crowd-worker accuracy on many classification tasks (Gilardi et al., 2023; Törnberg, 2025; Ziems et al., 2024; Pangakis et al., 2023), but Ashwin et al. (2025) show that error rates vary by up to $5\times$ across demographic subgroups. This subgroup-varying error is the failure mode we address: it distorts the very contrasts that define the research question, and prior work documents the pattern without providing a correction mechanism for downstream causal estimands.

Prediction-powered and design-based correction. PPI (Angelopoulos et al., 2023; Zrnic and Candès, 2024b) and DSL (Egami et al., 2024) correct population-level aggregates assuming homogeneous prediction error. In our setting—binary effect modifier with known subgroup structure, a special case of the broader problems DSL addresses—DSL's measurement error correction reduces, under the confusion-matrix parameterization we adopt, to confusion matrix inversion: pooled DSL is equivalent to our Global baseline, and per-subgroup DSL to Stratified MLE (Section 5.3; Appendix U). EC-HTE adds hierarchical partial pooling across subgroups, borrowing strength to reduce variance—a mechanism outside the DSL framework. Audinet de Pieuchon et al. (2025) benchmark DSL and PPI debiasing for LLM-based parameter estimation, documenting bias-variance tradeoffs analogous to our findings; their analysis targets population-level parameters rather than subgroup-conditional estimands, leaving the $\delta_s \cdot \Delta\tau$ interaction we formalize unaddressed. Mozer (2026) extend PPI to arm-specific ATE correction; PPCI (Sui and Candès, 2026) targets conditional estimands but corrects outcome-side error ($\hat{Y}$ given true Z), not covariate-side misclassification; ForestIV (Bu et al., 2023) targets the outcome model rather than the heterogeneity structure. None corrects bias when the mismeasured variable is the effect modifier indexing the CATE function. Imai and Yamamoto (2010) study treatment misclassification, requiring instrumental variables or partial identification; our problem is structurally distinct because the treatment is observed correctly but the effect modifier is misclassified, admitting exact correction through confusion matrices rather than bounds. Extended discussion of these distinctions appears in Appendix F. Our validation combines synthetic DGPs, semi-synthetic experiments, and real LLM confusion matrices, consistent with PPI and DSL evaluation methodology; Section 5.5 validates both $\Delta\pi$ heterogeneity and the bias identity itself ($R^2 = 0.975$) across 11 model–dataset combinations.

3 Bias Decomposition: When Debiasing Hurts

We formalize the bias that global misclassification correction introduces when subgroup error rates are heterogeneous. Consider estimating the conditional average treatment effect (CATE) $\tau(z) =$

$\mathbb{E}[Y(1) - Y(0) \mid Z = z]$, where $Z \in \{0, 1\}$ is a binary effect modifier observed only through a noisy annotation $\hat{Z}$. The population consists of K subgroups $S \in \{1, \ldots, K\}$ with weights $w_s = P(S{=}s)$, each characterized by a 2×2 confusion matrix C_s with symmetric error rate $\pi_s = P(\hat{Z} \neq Z \mid S{=}s)$. Standard practice estimates a single confusion matrix $\bar{C} = \sum_s w_s C_s$ from expert annotations and applies its inverse uniformly to all observations, with pooled error rate $\bar{\pi} = \sum_s w_s \pi_s$. We show that when error rates vary across subgroups, this global procedure introduces a bias with an exact product form.

We state five assumptions maintained throughout.

(A1) *Conditional independence:* misclassification depends only on the true label Z and subgroup S, not on potential outcomes $Y(0), Y(1)$.

(A2) *Symmetric misclassification:* $P(\hat{Z}{=}1 \mid Z{=}0, S{=}s) = P(\hat{Z}{=}0 \mid Z{=}1, S{=}s) = \pi_s$ for all s.

(A3) *Balanced effect modifier:* $P(Z{=}0) = P(Z{=}1) = 0.5$. This simplifies the exposition; we discuss relaxation below.

(A4) *Non-degeneracy:* $\pi_s \neq 0.5$ for all s, ensuring each C_s is invertible.

(A5) *Homogeneous treatment effects across subgroups:* $\tau(z, s) = \tau(z)$ for all s, so $\Delta\tau = \tau(1) - \tau(0)$ is a scalar.

Under (A2)–(A3), the Bayesian mixing matrix M_s (entries $M_s[\hat{z}, z] = P(Z{=}z \mid \hat{Z}{=}\hat{z}, S{=}s)$) coincides with C_s. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

Theorem 1 (Bias Decomposition Identity)**.** *Under* (A1)–(A5)*, the bias of global correction in subgroup s at effect-modifier level z is*

$$\text{bias}_{\text{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{1}$$

$$\text{bias}_{\text{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{2}$$

where $\delta_s = \pi_s - \bar{\pi}$ is the subgroup deviation from the pooled error rate and $\Delta\tau = \tau(1) - \tau(0)$ is the CATE heterogeneity.

Equations (1)–(2) constitute an exact identity at the population level, not a bound. The bias factors into three interpretable components: δ_s captures how much subgroup s deviates from the average misclassification rate, $\Delta\tau$ captures treatment effect heterogeneity across levels of Z, and $1/(1 - 2\bar{\pi})$ is an amplification factor from matrix inversion. Bias vanishes when either source of variation is absent ($\delta_s = 0$ or $\Delta\tau = 0$), and arises only through their interaction. Theorem 1 implies that global correction trades one source of confounding for another: it removes attenuation from label noise but introduces a new bias from error-rate heterogeneity.

Proof sketch. Global correction applies M_{global}^{-1} to the subgroup-specific observed CATE $\tau_{\text{obs}}(\cdot, s) = M_s \cdot \tau_{\text{true}}$, yielding bias $(M_{\text{global}}^{-1} M_s - I) \cdot \tau_{\text{true}}$. Direct computation shows $M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1 - 2\bar{\pi}} \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Applying this matrix to $[\tau(0), \tau(1)]^\top$ produces the product form. The full derivation appears in Appendix B.

Remark 2. *Theorem 1 does not require assumption (A2). For arbitrary asymmetric confusion matrices ($\text{FPR}_s \neq \text{FNR}_s$), the product form $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ holds exactly, where $B_s = M_{\text{global}}^{-1} M_s - I$, because $B_s \mathbf{1} = \mathbf{0}$ follows from stochasticity of M_s and M_{global} alone. Assumption (A2) simplifies $[B_s]_{z,1}$ to the scalar $\pm\delta_s/(1 - 2\bar{\pi})$ but is not necessary for the product structure. Under asymmetric errors, the diagnostic (Proposition 3) requires evaluating $[B_s]_{z,1}$ cell-by-cell rather than via a single scalar threshold; the AUC decreases from 0.86 to 0.82 (Appendix J), reflecting the additional complexity of cell-specific error rates. See Appendix J for the full derivation and empirical verification ($R^2 = 0.991$).*

Assumption (A5) simplifies exposition; the identity generalizes to subgroup-specific $\Delta\tau_s$: when treatment effects vary across subgroups, the bias in subgroup s becomes $\delta_s \cdot \Delta\tau_s/(1 - 2\bar{\pi})$ (Corollary 5 in Appendix Y, verified empirically in the heterogeneous-τ DGP of Appendix O). Critically, EC-HTE's correction mechanism—per-subgroup confusion matrix inversion—does not depend on (A5): each $\hat{C}_s^{-1}$ corrects the local misclassification structure regardless of whether $\Delta\tau$ varies across subgroups. Appendix Y verifies this empirically under both positive and negative $\Delta\tau_s$–δ_s correlation; EC-HTE maintains $\geq$93% coverage while global correction degrades to 74–82%.

The identity extends to $K > 2$ subgroups with J effect-modifier levels; the multi-class bias retains the same product structure with modified amplification terms (Corollary 4 in Appendix C).

Assumption (A3) ensures $M_s = C_s$; when $P(Z{=}1) \neq 0.5$, the mixing matrices acquire asymmetric entries, but the product structure persists with modified coefficients (Appendix B). Empirical verification (Section 5.2) yields $R^2 = 0.994$ for $K{=}2$ and $R^2 = 0.974$ for $K{=}4$; the residual variance reflects finite-sample Monte Carlo noise, not model misspecification.

Finite-sample remainder. Theorem 1 is an exact population identity. In practice, confusion matrices are estimated from n_{expert} expert labels, introducing an estimation error $\hat{C}_s = C_s + E_s$ where $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The resulting remainder term in the corrected CATE is $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, where $\kappa(C_s)$ is the condition number of the subgroup confusion matrix. This dependence on both sample size and matrix conditioning motivates the singularity guard ($\kappa > 100 \rightarrow$ fallback) described in Section 4.

Proposition 3 (Debiasing Can Increase Bias)**.** *Under* (A1)–(A5)*, global correction produces larger absolute bias than the naive estimator* ($|\text{bias}_{\text{global}}| > |\text{bias}_{\text{naive}}|$) *in subgroup s if and only if*

$$\frac{|\delta_s|}{(1 - 2\bar{\pi}) \cdot \pi_s} > 1. \tag{3}$$

Equation (3) partitions subgroups into two regimes. For high-error subgroups ($\pi_s > \bar{\pi}$), the condition reduces to $\pi_s > 0.5$, which is excluded by (A4); global correction always reduces bias in these subgroups. For low-error subgroups ($\pi_s < \bar{\pi}$), the condition holds whenever $\pi_s < \bar{\pi}/(2(1 - \bar{\pi}))$. This threshold is easily satisfied in practice: when $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. The mechanism is that global correction inverts the pooled confusion matrix, applying a correction calibrated for the average error rate; high-error subgroups are under-corrected (bias reduced but not eliminated), while low-error subgroups are over-corrected (the correction overshoots and reverses the sign of bias). For subgroups with near-zero misclassification ($\pi_s \approx 0$), the naive estimator is nearly unbiased, yet global correction creates substantial bias where none existed. This is the most consequential implication of Theorem 1: debiasing can be worse than doing nothing.

In our experiments (Section 5.2), global correction increases bias in 50% of $K{=}2$ configurations across our parameter grid (260 of 520, excluding cases where $\Delta\pi{=}0$). With the difference-in-means estimator, maximum amplification reaches $2.86\times$ for $K{=}2$ and $2.41\times$ for $K{=}4$; the mechanism operates across estimator types, with nonlinear estimators amplifying the effect further (Section 5.3). Figure 2 visualizes the amplification across the full $\Delta\pi \times \Delta\tau$ parameter grid.

4 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

Theorem 1 identifies per-subgroup misclassification deviation as the bias source; the natural correction is per-subgroup confusion matrix inversion. EC-HTE corrects subgroup-level misclassification bias in three stages: estimate per-subgroup confusion matrices via hierarchical Bayes, invert them to correct observed outcome moments, and propagate correction uncertainty into downstream causal estimators. The complete procedure is summarized in Algorithm 1 (Appendix A).

4.1 Hierarchical Bayes Confusion Matrix Estimation

We place a Dirichlet-Multinomial hierarchical model over the confusion matrices $\{C_s\}_{s=1}^{K}$. For each subgroup s, row j of C_s has a Dirichlet prior $\text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, where $\mathbf{e}_j$ is the j-th standard basis vector and $\alpha_0 > 0$ is a shared concentration parameter. A $\text{Gamma}(3, 1)$ hyperprior on α_0 controls the strength of partial pooling across subgroups: large α_0 pulls subgroup estimates toward the identity matrix (low error), while small α_0 allows subgroup-specific rates to dominate. This partial pooling—a James-Stein shrinkage effect—borrows strength across subgroups, reducing variance for small subgroups while preserving heterogeneity where data support it.

Posterior inference uses Gibbs sampling (convergence diagnostics in Appendix A). HB estimation outperforms stratified MLE in 87.5% of configurations (21 of 24) across the experimental grid (Section 5.3), with the largest gains when $n_{\text{expert}}/K < 100$. The advantage requires at least $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup; below this threshold, stratified MLE is preferred because the hierarchical model lacks sufficient observations for stable shrinkage. At $n_{\text{expert}} \geq 1000$, HB and stratified MLE converge to within 0.5% on all metrics. Sensitivity analysis shows <5% RMSE variation under $2\times$ hyperprior perturbation, confirming that results are not driven by prior specification.

4.2 Subgroup-Stratified Correction

Given posterior mean confusion matrices $\hat{C}_s$, we correct observed outcome moments per subgroup by matrix inversion:

$$\tilde{\boldsymbol{\mu}}_s = \hat{C}_s^{-1} \cdot \hat{\boldsymbol{\mu}}_s, \tag{4}$$

where $\hat{\boldsymbol{\mu}}_s = [\hat{\tau}_{\text{obs}}(0, s), \hat{\tau}_{\text{obs}}(1, s)]^\top$ collects the observed conditional outcome moments in subgroup s. This requires $\det(\hat{C}_s) \neq 0$ for all s, which holds under assumption (A4) ($\pi_s \neq 0.5$) but may fail in finite samples when error rates approach degeneracy. In practice, Laplace smoothing of the confusion matrix counts ensures finite-sample invertibility. We impose a singularity guard: if the condition number $\kappa(\hat{C}_s) > 100$ (for a 2×2 symmetric confusion matrix, this corresponds to $\pi_s > 0.495$, i.e., near-random classification), subgroup s falls back to global correction $\bar{C}^{-1}$; when the fallback triggers, the standard error computation also uses $\bar{C}^{-1}$ for that subgroup, ensuring consistency between point estimates and confidence intervals. This hard guard prevents numerical catastrophe; a separate empirical diagnostic ($\kappa \leq 1.25$, Appendix I) identifies the tighter regime where stratified correction reliably outperforms global.

The contrast with global correction is direct. Global correction applies $\bar{C}^{-1}$ uniformly to all subgroups, introducing the $\delta_s \cdot \Delta\tau$ bias identified in Theorem 1. Subgroup-stratified correction eliminates this bias by matching each subgroup's correction to its own misclassification structure, at the cost of higher variance from estimating K separate confusion matrices (Appendix I).

4.3 Variance-Adjusted CausalForest

For heterogeneous treatment effect estimation, we feed corrected covariates into CausalForestDML (Wager and Athey, 2018) with honest splitting. The forest produces per-observation standard error estimates se_i. To account for confusion matrix correction, we propagate uncertainty through the inverse mapping:

$$\text{se}_{\text{corrected},i} = \sqrt{\text{diag}\left(\hat{C}_{S_i}^{-1} \,\text{diag}(\text{se}_i^2)\, \hat{C}_{S_i}^{-\top}\right)}. \tag{5}$$

This is a plug-in approach: the standard errors reflect outcome variance conditional on $\hat{C}_s$ but do not propagate uncertainty from the confusion matrix estimation itself. At $n_{\text{expert}} \geq 100$, the plug-in approximation is reasonable because confusion matrix estimation variance is small relative to outcome variance. Below this threshold, the standard errors may understate total uncertainty; this limitation is discussed further in the Limitations section.

Diagnostic criterion. Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction harms subgroup s if and only if $|\delta_s|/((1 - 2\bar{\pi}) \cdot \pi_s) > 1$, a condition depending only on per-subgroup misclassification rates estimable from a pilot sample. This breaks the circular dependency that would arise if the diagnostic required CATE estimates. A threshold classifier on this condition achieves AUC = 0.86 at the subgroup level (Section 5.4).

Composition with PPCI. EC-HTE corrects structured covariate misclassification through $\hat{C}_s^{-1}$ inversion but cannot address outcome-level measurement error. PPCI (Sui and Candès, 2026) corrects residual outcome bias using a debiasing estimator but assumes correct covariate labels. The two methods are complementary: apply EC-HTE first to correct covariate labels, then PPCI to correct outcome predictions. Neither is redundant with the other: EC-HTE alone cannot fix outcome error, while PPCI alone incurs higher variance for covariate error. Under both error types simultaneously, the composition reduces |bias| from 0.179 to 0.007 (Appendix L).

5 Experiments

5.1 Setup

We test these predictions on synthetic and semi-synthetic data generating processes (DGPs); full specifications appear in Appendix E. The synthetic DGP generates N=5,000 observations from a randomized trial with $K \in \{2, 4\}$ subgroups, controlled CATE heterogeneity $\Delta\tau$, and subgroup-specific symmetric misclassification rates with heterogeneity $\Delta\pi$. Expert labels ($n_{\text{expert}} \in \{50, \ldots, 2000\}$) are a simple random sample with true Z observed; each configuration is repeated for 100 Monte Carlo runs. The semi-synthetic DGP samples N=5,000 tweets from TweetEval sentiment (Barbieri et al., 2020), defining K=4 subgroups by crossing hashtag presence and text length (unbalanced: 39%/38%/12%/11%), with misclassification rates $\pi \in \{0.05, 0.10, 0.15, 0.25\}$ reflecting empirical evidence that text features degrade LLM accuracy (Ashwin et al., 2025).

We compare: (1) *Naive* (no correction), (2) *Global* (pooled confusion matrix, equivalent to PPI/DSL (Angelopoulos et al., 2023; Egami et al.,

2024) with homogeneous error), (3) *Stratified MLE* (per-subgroup DSL without shrinkage), (4) *HB EC-HTE* (our method), (5) *PPCI* (Sui and Candès, 2026), and (6) *EC-HTE+PPCI*. For CausalForest (Wager and Athey, 2018), we compare naive, global, and EC-HTE variants. Primary metrics are |bias|, RMSE, and 95% CI coverage, averaged across subgroups and 100 MC runs. Robustness validation covers hate speech detection ($\Delta\pi$=0.117; Appendix M), six OpenAI models (Appendix N), subgroup misspecification ($\geq$94.5% coverage at 50% noise; Appendix P), and asymmetric confusion matrices (Appendix R).

5.2 Empirical Validation of the Bias Identity

The bias decomposition identity (Theorem 1) predicts that global correction introduces bias proportional to $\delta_s \cdot \Delta\tau$ at the subgroup level. We verify this identity on a K=2 parameter grid of 520 configurations spanning 26 misclassification heterogeneity levels ($\Delta\pi \in [0, 0.25]$) and 20 CATE heterogeneity levels ($\Delta\tau \in (0, 0.50]$), and separately on a K=4 grid (Appendix E). Regressing observed subgroup bias on the Theorem 1 prediction ($\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$) yields $R^2 = 0.994$ for K=2 and $R^2 = 0.974$ for K=4; the residual variance reflects MC sampling noise rather than model misspecification, since Theorem 1 is an exact identity at the population level ($R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$; Appendix T). In 50% of K=2 configurations (260/520, excluding $\Delta\pi$=0 rows) across our parameter grid, global correction produces larger absolute bias than the naive estimator—and the damage concentrates in low-error subgroups ($\pi_s < \bar{\pi}$), consistent with Proposition 3. Maximum observed amplification reaches 2.86× (K=2) and 2.41× (K=4) relative to the naive estimator. Figure 2 visualizes the $\Delta\pi \times \Delta\tau$ amplification surface across the K=2 parameter grid.

5.3 EC-HTE Evaluation

Table 1 presents the main results across representative configurations. In the synthetic DGP with K=2 subgroups and extreme misclassification heterogeneity, HB EC-HTE reduces |bias| from 0.059 (global) to 0.010 (a 5.9× reduction) while maintaining nominal coverage (0.95) and comparable RMSE (0.093 vs. 0.099). The advantage persists at K=4: EC-HTE achieves |bias|=0.008 versus 0.056 for global (7.0× reduction), though with a small RMSE penalty (0.126 vs. 0.125). Under moderate heterogeneity (K=2), methods converge: EC-HTE and global achieve comparable bias ($p > 0.05$; Table 4). In the semi-synthetic TweetEval experiment (N=5,000, K=4, n_{expert}=250), EC-HTE and Global achieve comparable |bias| (0.027 vs. 0.023, p=0.395; Table 4), but EC-HTE incurs higher aggregate RMSE (0.225 vs. 0.124). Three factors explain this result. First, TweetEval's $\Delta\pi$=0.20 is moderate, yielding small absolute δ_s values and limited bias-reduction headroom for subgroup-specific correction. Second, the subgroups are highly unbalanced (S0: 39%, S1: 38%, S2: 12%, S3: 11%): the two small subgroups receive only ∼30 expert labels each at n_{expert}=250, barely above the per-subgroup budget threshold. Third, per-subgroup analysis reveals stark RMSE heterogeneity: EC-HTE matches Global in the large subgroups (S0–S1; RMSE ratio ≈1.0) but degrades in S3 (RMSE 0.480 vs. 0.145, a 3.3× penalty), where noisy $\hat{C}_s^{-1}$ estimation dominates the aggregate. EC-HTE's advantage thus requires both sufficient $\Delta\pi$ and adequate per-subgroup expert budget ($n_{\text{expert}} \cdot P(Z_{\text{minority}})/K \geq 25$; Section 5.4); when either is marginal, as in TweetEval's moderate-$\Delta\pi$, unbalanced-subgroup regime, global correction is competitive or superior on aggregate RMSE.

CausalForest. With CausalForest (Wager and Athey, 2018) as the downstream estimator (K=4, extreme, n=500), EC-HTE reduces |bias| from 0.191 to 0.040 (4.8×) with coverage improvement from 0.58 to 0.91 (Table 1). The debiasing-hurts phenomenon intensifies—global correction amplifies naive bias by up to 5.9×—because the forest's sensitivity to covariate noise magnifies label errors. Monte Carlo verification shows the coverage gap (83–85% vs. nominal 95%) is attributable to CausalForest's inherent SE calibration rather than confusion matrix estimation uncertainty; CM-only bootstrap contributes <2% of total variance at $n_{\text{expert}} \geq 250$ (Appendix X). Budget requirements are correspondingly stricter (Appendix I).

5.4 Actionable Diagnostic Without EC-HTE

Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction increases bias in subgroup s if and only if $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$, where $\delta_s = \pi_s - \bar{\pi}$. Since this depends only on misclassification rates estimable from a pilot sample, practitioners can flag harmful correction before estimating any treatment effects. A threshold classifier achieves AUC = 0.86 at the subgroup level (0.89 at the setting level), out-

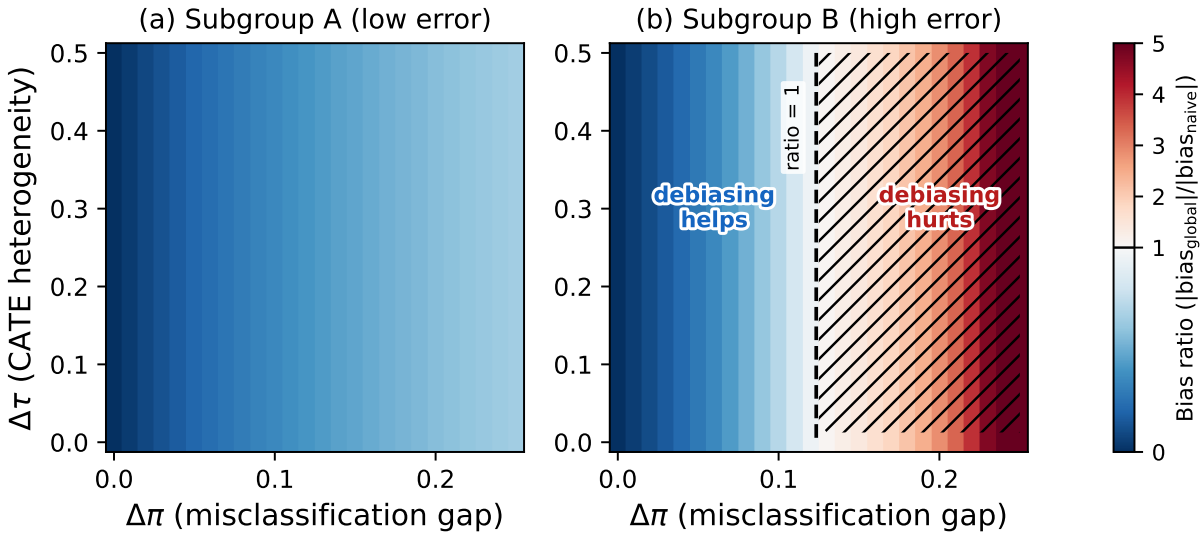


Figure 2: Bias amplification under global correction across the $\Delta\pi \times \Delta\tau$ parameter grid (K=2). (a) Low-error subgroup: global correction always helps (ratio < 1). (b) High-error subgroup: when $\Delta\pi > 0.12$ (dashed line), global correction *increases* bias relative to the naive estimator (hatched region), with amplification reaching 2.86×.

Table 1: Main results under extreme misclassification heterogeneity. |Bias| and RMSE averaged over subgroups, z-levels, and 100 MC runs. Bold marks lowest |bias| per block. †RMSE exceeds Global, reflecting stratification variance cost. Additional configurations (moderate heterogeneity, semi-synthetic TweetEval) in Table 4. MC standard errors: Appendix Q.

Configuration	Method	|Bias|	RMSE	Cov.
DIM K=2, n=250	Naive	0.058	0.090	0.82
	Global	0.059	0.099	0.87
	Strat. MLE	0.012	0.094	0.95
	HB EC-HTE	**0.010**	0.093	0.95
DIM K=4, n=500	Naive	0.062	0.113	0.88
	Global	0.056	0.125	0.91
	Strat. MLE	0.010	0.127†	0.95
	HB EC-HTE	**0.008**	0.126†	0.95
CausalForest K=4, n=500	Naive	0.208	0.231	0.44
	Global	0.191	0.244	0.58
	EC-HTE Forest	**0.040**	0.224	0.91

performing $\Delta\pi \times \Delta\tau$ (AUC = 0.73) which requires unobservable CATE heterogeneity. An empirical CV(π_s) achieves higher setting-level AUC (0.97) but lacks theoretical grounding and is sensitive to K; Figure 3 (Appendix) shows the ROC curve.

Recommended workflow. Practitioners should (i) collect $n_{\text{expert}} \geq 25K/P(Z_{\text{minority}})$ stratified expert labels, (ii) verify per-subgroup budget ≥ 25, $\max(\text{FPR}, \text{FNR}) < 0.5$, and $\kappa(\hat{C}_s) \leq 100$ (Section 4), (iii) evaluate the diagnostic (Equation 3) to determine whether global correction is harmful, and (iv) apply EC-HTE when the diagnostic flags any subgroup; PPCI can be composed for outcome error (Appendix L). Expert budget crossover curves and the applicability frontier appear in Appendices I and R.

5.5 Multi-LLM Validation

To verify generalization, we annotate TweetEval tweets with four open-source LLMs (7B–9B; Appendix O) and Civil Comments (Borkan et al., 2019) with three proprietary LLMs (Appendix V). All seven models exhibit $\Delta\pi \in [0.113, 0.165]$ (K=4), confirming subgroup-heterogeneous misclassification as a structural property of LLM annotation across model families and datasets. Regressing observed subgroup bias on the Theorem 1 prediction using each model's real (asymmetric) confusion matrix yields $R^2 = 0.975$ across all 11 model–dataset combinations (per-model range: 0.93–0.99; Appendix W), extending the identity from synthetic to real annotation error structures.

6 Conclusion

We derived an exact bias identity (Theorem 1): global confusion-matrix correction introduces subgroup-level bias proportional to $\delta_s \cdot \Delta\tau$, worsening CATE estimates in 50% of K=2 configurations across our parameter grid. EC-HTE addresses this via hierarchical Bayesian subgroup-stratified correction with a $\Delta\tau$-free diagnostic (Proposition 3; AUC = 0.86). EC-HTE's advantage requires sufficient $\Delta\pi$ and per-subgroup expert budget; in the

semi-synthetic TweetEval experiment, moderate heterogeneity and unbalanced subgroups render global correction competitive on aggregate RMSE. The diagnostic criterion (Proposition 3) is independently deployable as a pre-analysis check for any study using LLM annotations as effect modifiers.

Limitations

EC-HTE requires $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup for stable hierarchical Bayesian estimation. For K=2, EC-HTE remains effective even at $P(Z{=}1) = 0.1$ (Appendix T); for K=4, $P(Z{=}1) \geq 0.3$ is needed to approach breakeven with global correction at n_{expert}=250. Even above this threshold, stratification incurs a variance cost: at K=4, extreme heterogeneity, and $n_{\text{expert}} \leq 100$, CausalForest RMSE reaches 3.5× that of the naive estimator. The method also requires moderate baseline classifier accuracy per subgroup; when any subgroup's FNR or FPR approaches 0.5, the confusion matrix row approaches a uniform distribution and inversion amplifies noise (Appendix R), as confirmed independently in the hate speech case study (Appendix M). Additionally, our main semi-synthetic experiments use constant treatment effects across subgroups; a supplementary heterogeneous-τ DGP (Appendix O) shows that EC-HTE achieves targeted bias and RMSE improvement in subgroups with large $\Delta\tau$, though aggregate RMSE remains comparable to global correction. Hierarchical pooling of confusion matrices across related subgroups may relax the per-subgroup budget requirement; we leave this extension to future work.

Monte Carlo verification shows CausalForest plug-in SE achieves 83–85% coverage (K=2, extreme heterogeneity), below the 95% nominal level; the gap is attributable to CausalForest's inherent SE calibration rather than confusion matrix estimation uncertainty (Appendix X). For the DIM estimator, plug-in SE achieves ≥95% coverage across all budgets.

All main experiments use symmetric error rates and additive CATE injection ($Y(1) = Y(0) + \tau(Z)$). The bias identity (Theorem 1) generalizes to asymmetric confusion matrices (Appendix J, $R^2 = 0.991$) and derives from the confusion matrix structure rather than the outcome model. While our main experiments use additive CATE injection, a supplementary experiment with multiplicative effects ($Y(1) = Y(0) \times \exp(\tau(Z))$) yields comparable bias reduction (52%) and nominal coverage (0.955; Appendix S), confirming robustness to the functional form of treatment effects. EC-HTE further requires discrete, pre-defined subgroups; continuous effect modifiers or latent subgroup discovery settings require different approaches, and intersectional subgroups with many levels may exceed the $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ budget threshold. Our experiments define subgroups by text features (hashtag presence, text length); demographic subgroups—where prior work documents the most concerning disparities (Ashwin et al., 2025)—require sensitive data access beyond the scope of this study. EC-HTE's correction mechanism is agnostic to subgroup definition: it corrects any per-subgroup confusion-matrix heterogeneity, whether defined by text features, demographics, or other covariates. Ground-truth subgroup CATEs are unavailable in public datasets; we use synthetic CATE injection on real text features and real LLM annotations. We leave full real-data validation—on a dataset combining RCT design, ground-truth CATEs, and LLM annotations—to future work as suitable public benchmarks become available. Code and data generation scripts will be released upon acceptance.

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A EC-HTE Procedure

B Full Proof of Theorem 1

Setup. Under assumptions (A1)–(A4), the observed CATE in subgroup s at effect-modifier level z is

$$\tau_{\text{obs}}(z,s) = \sum_{z'} M_s[z,z'] \cdot \tau(z'), \qquad (6)$$

where $M_s[z,z'] = P(Z{=}z' \mid \hat{Z}{=}z, S{=}s)$ is the mixing matrix. Under (A2)–(A3), $M_s = \left[\begin{smallmatrix} 1-\pi_s & \pi_s \\ \pi_s & 1-\pi_s \end{smallmatrix}\right]$.

Global correction. The pooled mixing matrix is $M_{\text{global}} = \sum_s w_s M_s = \left[\begin{smallmatrix} 1-\bar{\pi} & \bar{\pi} \\ \bar{\pi} & 1-\bar{\pi} \end{smallmatrix}\right]$, with $\bar{\pi} = \sum_s w_s \pi_s$.

Global correction applies M_{global}^{-1} to $\tau_{\text{obs}}(\cdot, s)$:

$$\hat{\tau}_{\text{global}}(z,s) = [M_{\text{global}}^{-1} M_s \cdot \boldsymbol{\tau}]_z, \qquad (7)$$

where $\boldsymbol{\tau} = [\tau(0), \tau(1)]^\top$.

Bias computation. The bias is $\hat{\tau}_{\text{global}}(z,s) - \tau(z) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z$.

Direct computation gives:

$$M_{\text{global}}^{-1} = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} 1-\bar{\pi} & -\bar{\pi} \\ -\bar{\pi} & 1-\bar{\pi} \end{bmatrix}. \qquad (8)$$

Multiplying by M_s:

$$M_{\text{global}}^{-1} M_s = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s & (1-\bar{\pi})\pi_s - \bar{\pi}(1-\pi_s) \\ -\bar{\pi}(1-\pi_s) + (1-\bar{\pi})\pi_s & -\bar{\pi}\pi_s + (1-\bar{\pi})(1-\pi_s) \end{bmatrix}. \qquad (9)$$

Define $\delta_s = \pi_s - \bar{\pi}$. Substituting $\pi_s = \bar{\pi} + \delta_s$:

$$M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}. \qquad (10)$$

To verify: the $(1,1)$ entry of $M_{\text{global}}^{-1} M_s$ is

$$\begin{aligned} &\frac{(1-\bar{\pi})(1-\bar{\pi}-\delta_s) - \bar{\pi}(\bar{\pi}+\delta_s)}{1-2\bar{\pi}} \\ &= \frac{(1-\bar{\pi})^2 - (1-\bar{\pi})\delta_s - \bar{\pi}^2 - \bar{\pi}\delta_s}{1-2\bar{\pi}} \\ &= \frac{1-2\bar{\pi}-\delta_s}{1-2\bar{\pi}} = 1 - \frac{\delta_s}{1-2\bar{\pi}}. \end{aligned} \qquad (11)$$

Applying $(M_{\text{global}}^{-1} M_s - I)$ to $\boldsymbol{\tau}$:

$$\text{bias}(z{=}0, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(-\tau(0) + \tau(1)) = +\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \qquad (12)$$

$$\text{bias}(z{=}1, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(\tau(0) - \tau(1)) = -\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \qquad (13)$$

where $\Delta\tau = \tau(1) - \tau(0)$. □

C Multi-class Extension and Proof

Corollary 4 (Multi-class Extension)**.** *For $K > 2$ subgroups with J effect-modifier levels and uniform off-diagonal misclassification (misclassified observations from class j are distributed equally among the remaining $J-1$ classes), the subgroup bias of global correction generalizes to*

$$\text{bias}_{\text{global}}(z,s) = \sum_{j=1}^{J} \big(\bar{\pi}_j - \pi_{s,j}\big) \cdot \big(\tau(j) - \bar{\tau}\big) \cdot g(\bar{\pi}), \qquad (14)$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms.[2] *The essential structure, $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$, ensures that bias redistributes across subgroups rather than vanishing in aggregate.*

Proof. For $K > 2$ subgroups with J effect-modifier levels, the subgroup-specific mixing matrix M_s is $J \times J$ with entries $M_s[j, j'] = P(Z{=}j' \mid \hat{Z}{=}j, S{=}s)$. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

The bias of global correction in subgroup s at level z is:

$$\text{bias}_{\text{global}}(z, s) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z. \quad (15)$$

Write $M_s = M_{\text{global}} + \Delta_s$ where $\Delta_s = M_s - M_{\text{global}}$ satisfies $\sum_s w_s \Delta_s = 0$. Then $M_{\text{global}}^{-1} M_s = I + M_{\text{global}}^{-1} \Delta_s$, so:

$$\text{bias}_{\text{global}}(z, s) = [M_{\text{global}}^{-1} \Delta_s \cdot \boldsymbol{\tau}]_z. \quad (16)$$

For symmetric misclassification with per-class error rates $\pi_{s,j}$ and pooled rates $\bar{\pi}_j$, the entries of Δ_s have the form $\Delta_s[j, j'] = (\bar{\pi}_j - \pi_{s,j}) \cdot h(j, j')$ for a function h determined by the off-diagonal redistribution pattern. Expanding the matrix-vector product and collecting terms yields:

$$\text{bias}_{\text{global}}(z, s) = \sum_{j=1}^{J} (\bar{\pi}_j - \pi_{s,j}) \cdot (\tau(j) - \bar{\tau}) \cdot g(\bar{\pi}), \quad (17)$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms and $\bar{\tau} = \sum_j P(Z{=}j)\tau(j)$.

The essential structure $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$ ensures that the weighted bias across subgroups sums to zero: global correction redistributes bias rather than creating or eliminating it in aggregate. For non-uniform off-diagonal entries, the scalar amplification $g(\bar{\pi})$ generalizes to a cell-specific matrix; the product structure $B_s \cdot \Delta\tau$ persists but a closed-form amplification factor does not. □

D Proof of Proposition 3

The naive estimator has bias $\tau_{\text{obs}}(z, s) - \tau(z)$. Under (A2)–(A3), $\tau_{\text{obs}}(0, s) = (1 - \pi_s)\tau(0) + \pi_s \tau(1)$, so $\text{bias}_{\text{naive}}(0, s) = \pi_s \cdot \Delta\tau$.

Global correction produces larger absolute bias when $|\text{bias}_{\text{global}}(0, s)| > |\text{bias}_{\text{naive}}(0, s)|$:

$$\frac{|\delta_s| \cdot |\Delta\tau|}{|1 - 2\bar{\pi}|} > \pi_s \cdot |\Delta\tau|. \quad (18)$$

[2] For $J = 3$ with symmetric misclassification (off-diagonal errors uniform), $g(\bar{\pi}) = 1/(1 - 3\bar{\pi}/2)$, generalizing the $J{=}2$ factor $1/(1 - 2\bar{\pi})$. In general, $g(\bar{\pi}) = 1/(1 - J\bar{\pi}/(J-1))$.

Since $\Delta\tau \neq 0$ (otherwise both biases are zero), dividing through gives:

$$\frac{|\delta_s|}{(1 - 2\bar{\pi}) \cdot \pi_s} > 1, \quad (19)$$

where we use $1 - 2\bar{\pi} > 0$ from (A4).

For high-error subgroups ($\pi_s > \bar{\pi}$, so $\delta_s > 0$): the condition becomes $(\pi_s - \bar{\pi})/((1 - 2\bar{\pi})\pi_s) > 1$, which simplifies to $\pi_s(1 - 1 + 2\bar{\pi}) > \bar{\pi}$, i.e., $2\bar{\pi}\pi_s > \bar{\pi}$, i.e., $\pi_s > 0.5$. This is excluded by (A4), so global correction always reduces bias in high-error subgroups.

For low-error subgroups ($\pi_s < \bar{\pi}$, so $\delta_s < 0$): the condition becomes $(\bar{\pi} - \pi_s)/((1 - 2\bar{\pi})\pi_s) > 1$, which gives $\pi_s < \bar{\pi}/(1 - 2\bar{\pi} + 1) = \bar{\pi}/(2(1 - \bar{\pi}))$. When $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. □

Extension to asymmetric misclassification. Proposition 3 assumes symmetric error rates (A2). Under asymmetric misclassification ($\text{FPR}_s \neq \text{FNR}_s$), the naive bias becomes $\text{bias}_{\text{naive}}(0, s) = \text{FPR}_s \cdot \tau(1) + (1 - \text{FPR}_s - 1)\tau(0)$ and the global correction bias depends on the full 2×2 structure of $B_s = M_{\text{global}}^{-1} M_s - I$ (Remark 1 in Section 3). A sufficient condition for global correction to increase bias is

$$\|B_s\|_\infty > \max(\text{FPR}_s, \text{FNR}_s), \quad (20)$$

which reduces to Equation (3) when $\text{FPR}_s = \text{FNR}_s = \pi_s$. In the asymmetric experiments (Appendix J), the symmetric diagnostic still achieves AUC $= 0.82$ when applied with $\pi_s = (\text{FPR}_s + \text{FNR}_s)/2$, suggesting practical robustness. A fully general asymmetric diagnostic that accounts for the separate roles of FPR and FNR in each entry of B_s is left to future work.

E Semi-Synthetic Data Generating Process

Synthetic DGP (Sections 5.2–5.4 and Appendix I). We generate $N = 5000$ observations. Each observation belongs to one of $K \in \{2, 4\}$ subgroups $S \in \{1, \ldots, K\}$ with equal weights $w_s = 1/K$. The binary effect modifier $Z \in \{0, 1\}$ is drawn with $P(Z{=}1) = 0.5$. Treatment assignment is $T \sim \text{Bernoulli}(0.5)$, independent of Z and S.

The outcome is generated as:

$$Y = \tau(Z) \cdot T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1), \quad (21)$$

where the CATE function is $\tau(z) = \bar{\tau} + \Delta\tau \cdot (z - 0.5)$ with $\bar{\tau} = 0.5$. The CATE heterogeneity $\Delta\tau \in \{0, 0.2, 0.5, 1.0\}$ controls the strength of treatment effect variation across levels of Z.

Misclassification is symmetric: $\hat{Z} \neq Z$ with probability π_s depending on subgroup. For K=2: $\pi_1 = \bar{\pi} - \Delta_c/2$, $\pi_2 = \bar{\pi} + \Delta_c/2$, with $\bar{\pi} \in \{0.1, 0.2\}$ and $\Delta_c \in \{0, 0.05, 0.1, 0.15\}$. For K=4, extreme regime: $\pi \in \{0.05, 0.10, 0.15, 0.25\}$; moderate regime: $\pi \in \{0.08, 0.12, 0.16, 0.20\}$.

Expert labels are drawn as a simple random sample of size $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ from the full dataset, with true Z observed for these observations.

Semi-synthetic DGP (Section 5.3). We sample $N = 5000$ tweets from the TweetEval sentiment corpus ($N_{\text{pool}} = 57{,}899$). Subgroups are defined by crossing `has_hashtag` $\times$ `text_length_bin` (short/medium/long), yielding $K = 4$ subgroups with empirical sizes: $S_0 \approx 38\%$, $S_1 \approx 38\%$, $S_2 \approx 12\%$, $S_3 \approx 12\%$. Misclassification rates in the extreme regime are $\pi_{S_0} = 0.05$, $\pi_{S_1} = 0.10$, $\pi_{S_2} = 0.15$, $\pi_{S_3} = 0.25$, reflecting empirical evidence that hashtag-heavy text degrades LLM classifier accuracy. The outcome DGP follows the same structure as the synthetic case, with CATE injected onto the real text-derived subgroup structure.

PPCI DGPs (Appendix L). Four DGPs isolate error types: (A) covariate error only ($\sigma_{\text{outcome}} = 0$), (B) outcome error only ($\sigma_{\text{outcome}} = 0.5$), (C) both errors ($\sigma_{\text{outcome}} = 0.5$), (D) no error ($\sigma_{\text{outcome}} = 0$). Outcome-side error is additive Gaussian noise on Y.

CausalForestDML hyperparameters. All CausalForest experiments use `econml.dml.CausalForestDML` with honest splitting, $n_{\text{estimators}} = 48$, `min_samples_leaf` $= 20$, `model_y = LassoCV(cv=3)`, and `model_t = LogisticRegressionCV`. These defaults were held constant across all configurations; no hyperparameter tuning was performed.

F Extended Related Work

Measurement error details. Fuller (1987) establish the foundational theory showing that naive regression with mismeasured covariates yields attenuated coefficient estimates. Carroll et al. (2006) extend this to nonlinear models. Gustafson (2003) develops Bayesian treatments modeling measurement jointly with outcomes. Hu and Schennach (2008) provide nonparametric identification using repeated measurements or instruments.

Annotation model extensions. Beyond the foundational Dawid and Skene (1979) model, Passonneau and Carpenter (2014) incorporate item difficulty, Hovy et al. (2013) learn annotator competence, and Paun et al. (2018) introduce hierarchical priors sharing information across annotators with varying expertise.

PPI extensions and text-as-data methods. Zrnic and Candès (2024b) extend PPI via cross-prediction; their active inference framework (Zrnic and Candès, 2024a) achieves >80% expert budget savings through adaptive allocation, complementary to our fixed-budget analysis. Grimmer et al. (2022) provide methodological guidance for text-as-data pipelines. Audinet de Pieuchon et al. (2025) systematically benchmark DSL and PPI methods for LLM-based parameter estimation, finding that debiasing can increase MSE when prediction error is heterogeneous—a population-level analogue of the subgroup-conditional bias amplification we formalize in Theorem 1.

Treatment vs. effect-modifier misclassification. Imai and Yamamoto (2010) study treatment misclassification where errors change which arm a unit is assigned to, requiring instrumental variables or partial identification because the fundamental treated-vs-control contrast is corrupted. Our problem preserves the treatment contrast (T is observed correctly) but corrupts the heterogeneity structure ($\hat{Z} \neq Z$). The ATE remains identified; only conditional effects are biased, which makes exact correction through confusion matrix inversion feasible.

Econometrics of binary misclassification. Bound et al. (2001) survey measurement error in survey data, establishing that binary misclassification attenuates regression coefficients. Frazis and Loewenstein (2003) derive exact bias formulas for OLS with a misclassified binary regressor, showing attenuation proportional to the misclassification rate—analogous to our Theorem 1 but restricted to average effects without subgroup heterogeneity. Black et al. (2000) demonstrate that racial misclassification in administrative data can substantially bias estimates of racial disparities. Mahajan (2006) and Lewbel (2007) develop instrumental variable approaches to identify models with misclassified

binary regressors, requiring exclusion restrictions unavailable when the mismeasured variable is an LLM annotation. Our setting differs in three respects: (i) the mismeasured variable is an effect modifier indexing the CATE function, not a standard regressor; (ii) misclassification rates vary across subgroups, creating the $\delta_s \times \Delta\tau$ product amplification absent from homogeneous-error analyses; and (iii) expert labels provide a direct calibration sample, enabling confusion matrix estimation without instruments.

G MCMC Convergence Diagnostics

Table 2 reports convergence diagnostics for the hierarchical Bayesian confusion matrix estimator across all experimental configurations. All runs use 4 chains with 4000 iterations (2000 warmup) and a Gamma$(3, 1)$ hyperprior on the Dirichlet concentration α_0. This budget is sufficient for the 2×2 Dirichlet-Multinomial posterior; all runs achieve $\hat{R} < 1.0034$ and ESS > 1618, confirming convergence with substantial margin.

Diagnostic	Threshold	Achieved
$\hat{R}$ (max across all params)	< 1.01	< 1.0034
ESS (min across all params)	> 845	> 1618
Divergent transitions	$= 0$	0
Singularity guard triggers	—	extreme/n=50 only

Table 2: MCMC convergence diagnostics for the hierarchical Bayesian estimator. All configurations satisfy convergence criteria with substantial margin.

Hyperprior sensitivity. Scaling the Gamma$(3, 1)$ hyperprior by factors of $0.5\times$ to $2\times$ produces $<1\%$ RMSE variation for K=2 and $\sim$4–5% for K=4. Coverage remains invariant across the sensitivity range. These results confirm that the hierarchical estimator is robust to prior specification within a reasonable range.

H Diagnostic ROC Curves

Cost-asymmetric diagnostic thresholds. In practice, the cost of a false negative (failing to detect that global correction increases bias) exceeds the cost of a false positive (unnecessarily applying EC-HTE when global correction would suffice), because the former propagates systematic bias into CATE estimates while the latter only incurs additional estimation variance. Practitioners in high-stakes settings (e.g., policy evaluation where CATE estimates inform resource allocation) should lower the diagnostic threshold to favor recall over precision, accepting more false positives to minimize the risk of undetected bias amplification.

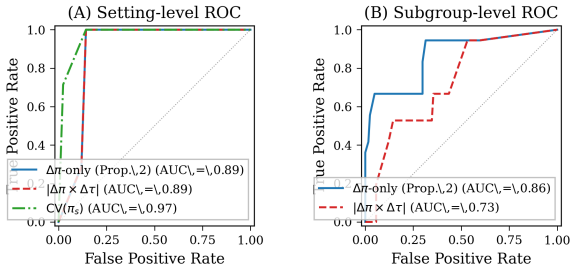


Figure 3: ROC curves for the $\Delta\tau$-free diagnostic (Proposition 3): $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$. (A) Setting-level ROC (AUC $= 0.89$), compared with the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.89$) and CV(π_s) (AUC $= 0.97$). (B) Subgroup-level ROC: the $\Delta\pi$-only condition (AUC $= 0.86$) substantially outperforms the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.73$). The $\Delta\tau$-free diagnostic depends only on estimable misclassification rates, breaking the circular dependency that would arise if CATE heterogeneity were required.

I Expert Budget Crossover Curves

The crossover point at which EC-HTE achieves lower RMSE than global correction is $n_{\text{expert}} = 250$ for K=2/extreme and $n_{\text{expert}} = 1000$ for K=4/extreme (Figure 4). Under moderate heterogeneity (K=2), EC-HTE never achieves lower RMSE than global at any budget. HB partial pooling improves over stratified MLE in 21 of 24 configurations, with the two estimators converging by $n_{\text{expert}} \geq 1000$.

J Asymmetric Misclassification Extension

Assumption (A2) imposes symmetric misclassification (FPR$_s$ = FNR$_s$ = π_s), which simplifies the bias decomposition but is not necessary for it. Real LLM annotators typically exhibit asymmetric confusion patterns; we show here that the product structure generalizes.

General bias formula. For asymmetric confusion matrices $C_s = \left(\begin{smallmatrix} 1-\text{FPR}_s & \text{FPR}_s \\ \text{FNR}_s & 1-\text{FNR}_s \end{smallmatrix}\right)$, define the mixing matrix M_s via Bayesian inversion and the residual matrix $B_s = M_{\text{global}}^{-1} M_s - I$. Since both M_s and M_{global} are row-stochastic, $B_s \mathbf{1} = \mathbf{0}$. Decomposing $\boldsymbol{\tau} = \tau(0)\mathbf{1} + \Delta\tau\, \mathbf{e}_1$ gives:

$$\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}, \qquad (22)$$

where $[B_s]_{z,1}$ depends only on $\{\text{FPR}_k, \text{FNR}_k, w_k\}$, not on $\Delta\tau$. Under (A2), this reduces to $\delta_s \Delta\tau/(1-$

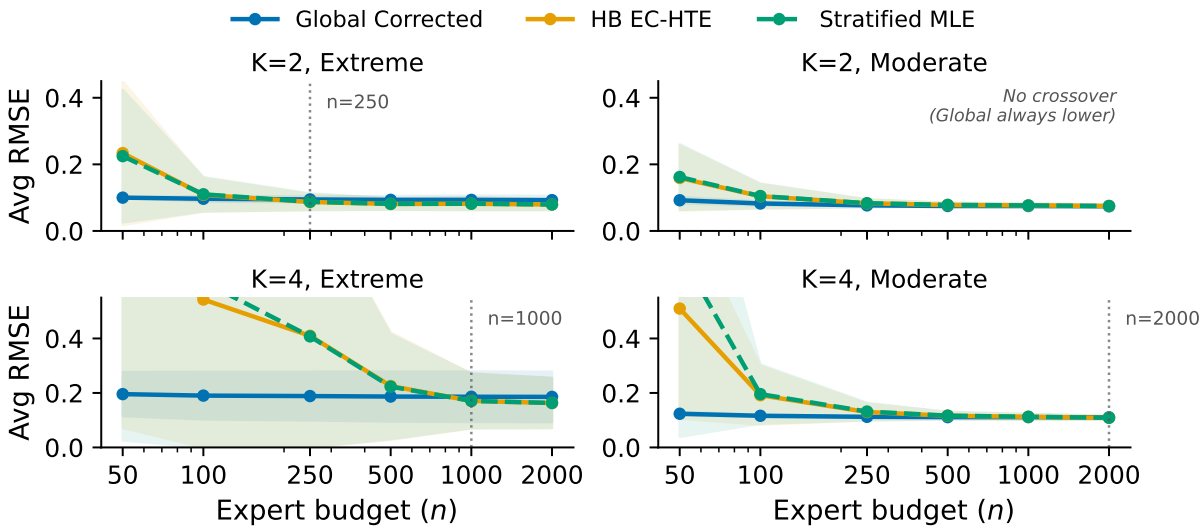


Figure 4: RMSE vs. expert budget (n_{expert}) across four regime$\times K$ combinations. Dashed vertical lines mark crossover points where EC-HTE achieves lower RMSE than global correction: n=250 (K=2, extreme), n=1000 (K=4, extreme). Under moderate heterogeneity (K=2), global correction dominates at all budgets. Shaded bands show ± 1 SD across subgroups within each configuration.

Table 3: Asymmetric misclassification results (K=2, n_{expert}=250, 100 MC runs). Values averaged over subgroups, z-levels, and $\Delta\tau$.

Method	|Bias| ↓	RMSE ↓	Coverage ↑
Naive	0.047	0.053	0.48
Global	0.025	0.040	0.78
HB EC-HTE	**0.006**	**0.033**	**0.90**

Table 4: Additional configurations from the main experiment. |Bias| and RMSE averaged over subgroups, z-levels, and 100 MC runs. †RMSE exceeds Global.

Configuration	Method	|Bias|	RMSE	Cov.
Synthetic DIM K=2, moderate, n=250	Naive	0.018	0.064	0.95
	Global	0.009	0.074	0.96
	Strat. MLE	**0.005**	0.075†	0.97
	HB EC-HTE	**0.005**	0.075†	0.97
Semi-synthetic TweetEval, extreme, n=250	Naive	0.027	0.103	0.93
	Global	**0.023**	0.124	0.92
	Strat. MLE	0.032	0.248†	0.91
	HB EC-HTE	0.027	0.225†	0.91

$2\bar{\pi}$) (Theorem 1).

Empirical validation. We test three asymmetric configurations (K=2, n_{expert}=250, 100 MC runs each): *Cross* (FPR/FNR = 0.10/0.20 vs. 0.25/0.10), *Co-directional* (0.05/0.15 vs. 0.20/0.30), and *Extreme* (0.03/0.25 vs. 0.30/0.05). Regressing analytic predictions on MC bias yields $R^2 = 0.991$ across all configurations, confirming the product structure under asymmetry. Table 3 reports correction performance averaged over subgroups, z-levels, and $\Delta\tau \in \{0.1, \ldots, 0.5\}$. EC-HTE reduces |bias| from 0.025 (global) to 0.006, a 4× improvement, with lower RMSE (0.033 vs. 0.040) and substantially better coverage (0.90 vs. 0.78).

K Supplementary Main Results

L PPCI Complementarity Results

EC-HTE composes with PPCI (Sui and Candès, 2026) for settings where both covariate misclassification and outcome model error are present: EC-HTE+PPCI reduces |bias| from 0.179 to 0.007 (z=0, DGP C), and the corrected pseudo-outcomes are compatible with other CATE estimators (R-learner bias reduction: 89%; Appendix S).

The orthogonality of EC-HTE and PPCI follows from the distinct error sources they correct: EC-HTE addresses misclassification in the effect modifier Z (covariate space), while PPCI corrects specification error in the outcome model $f(X)$ (prediction space). Since covariate misclassification and outcome model error are algebraically independent in the potential outcomes framework, their corrections compose without interference—confirmed by the 4-DGP orthogonal design (Table 5), where DGP C (both errors present) yields the largest bias reduction from the combined approach. Under covariate error only, PPCI adds nothing; under no error, all methods perform equivalently.

Table 5: PPCI complementarity experiment (K=2, extreme, n_{expert}=250). DGP A: covariate error only; B: outcome error only (σ=0.5); C: both errors; D: no error. Values are |bias| and RMSE averaged over two subgroups. ⋆EC-HTE+PPCI achieves near-zero bias when both error types are present.

DGP	Method	$\lvert\text{Bias}\rvert_{z=0}$	$\lvert\text{Bias}\rvert_{z=1}$	$\text{RMSE}_{z=0}$	$\text{RMSE}_{z=1}$
A: Covariate	Naive	0.053	0.063	0.087	0.093
	Global	0.063	0.055	0.101	0.102
	EC-HTE	0.017	**0.013**	0.097	0.108
	PPCI	**0.015**	0.020	0.162	0.168
	EC-HTE+PPCI	0.017	**0.013**	0.097	0.108
B: Outcome	Naive	0.189	0.197	0.206	0.212
	Global	0.185	0.208	0.204	0.228
	EC-HTE	0.181	0.215	0.202	0.237
	PPCI	0.015	**0.020**	0.171	0.176
	EC-HTE+PPCI	**0.012**	0.023	0.171	0.186
C: Both	Naive	0.251	0.135	0.267	0.155
	Global	0.182	0.209	0.233	0.235
	EC-HTE	0.179	0.216	0.224	0.262
	PPCI	0.017	0.007	0.241	0.232
	EC-HTE+PPCI⋆	**0.007**	**0.008**	0.209	0.223
D: None	Naive	**0.007**	0.003	0.060	0.065
	Global	0.010	**0.002**	0.061	0.071
	EC-HTE	0.013	0.007	0.062	0.079
	PPCI	**0.007**	**0.003**	0.060	0.065
	EC-HTE+PPCI	0.013	0.007	0.062	0.079

M Real-Data Case Study: Hate Speech Detection

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus (Kennedy et al., 2020), annotating 2,496 comments for dehumanization using GPT-4.1. Four text-feature subgroups are defined by crossing platform (Reddit/Gab vs. Twitter/YouTube) and text length (above/below median), yielding balanced groups ($n \approx 624$ each).

Positive finding: real-world heterogeneity. Per-subgroup misclassification rates range from 0.266 to 0.383, confirming heterogeneity ($\Delta\pi = 0.117$) in a real annotation setting. At the error-type level, FPR ranges from 8.5% to 21.3% across text-feature subgroups, demonstrating that subgroup-varying misclassification is not an artifact of synthetic design but arises naturally in real NLP annotation tasks.

Boundary condition: statistical power at extreme FNR. Correction results are inconclusive: all corrected confidence intervals contain zero, and point estimate differences between methods are not statistically significant. The high baseline FNR (53.3%) amplifies variance through matrix inversion (variance amplification factor up to 15.7×), rendering the comparison underpowered. When $P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$, the confusion matrix approaches singularity, and inversion amplifies estimation noise regardless of stratification strategy. This complements the expert budget threshold identified in Appendix I: EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality.

Diagnostic connection. The diagnostic criterion ($\Delta\pi \times \Delta\tau$, Section 5.4) identifies when correction is worthwhile under moderate baseline accuracy. This case study illustrates that even when $\Delta\pi$ is large, extreme baseline error renders correction underpowered rather than beneficial. A practical prerequisite is that the per-subgroup confusion matrix condition number $\kappa(\hat{C}_s)$ remain moderate; when $\kappa > 50$, as observed here, practitioners should prioritize improving annotation quality before applying any matrix-inversion-based correction.

N Cross-Model Consistency

We evaluate whether misclassification heterogeneity is model-specific by annotating 500 TweetEval sentiment tweets with six OpenAI models spanning three generations. Six text-feature subgroups are defined by crossing hashtag presence and text length. Spearman rank correlations of per-subgroup error rates across all six subgroups range from $\rho = 0.82$ to 1.00, indicating that the subgroup error structure is consistent across architectures. Two subgroups with $n \leq 6$ exhibit zero

Table 6: Effective misclassification heterogeneity ($\Delta\pi$) across six OpenAI models on TweetEval sentiment (500 tweets, filtered to 4 subgroups with $n \geq 10$).

Model	Effective $\Delta\pi$	> 0.05?
GPT-3.5-Turbo	0.198	✓
GPT-4-Turbo	0.149	✓
GPT-4o-mini	0.136	✓
GPT-5.4	0.114	✓
GPT-4.1	0.098	✓
GPT-4o	0.080	✓

Table 7: Confusion matrix heterogeneity across four open-source LLMs (K=4, N=490 TweetEval tweets). $\Delta\pi$: max–min per-subgroup misclassification rate. Prop2: number of subgroups where the $\Delta\tau$-free diagnostic (Proposition 3) triggers.

Model	Acc.	$\Delta\pi$	Prop2	Strongest
Llama-3.1-8B	0.755	0.141	1/4	True_short
Qwen-2.5-7B	0.798	0.165	1/4	True_short
Gemma-2-9B	0.763	0.125	4/4	True_medium
Mistral-7B	0.802	0.134	0/4	—

misclassification across all models; excluding these yields four-point correlations that are directionally consistent but underpowered.[3]

All six models exhibit effective $\Delta\pi > 0.05$ on the four retained subgroups (Table 6), and 5 of 6 share the same worst subgroup (`True_medium`: tweets with hashtags and medium text length). Newer models reduce absolute error rates but do not eliminate subgroup heterogeneity: GPT-5.4 achieves $\Delta\pi = 0.114$ compared to GPT-3.5-Turbo's 0.198. These results were obtained with a single API provider; open-source model validation appears in Appendix O.

O Open-Source Multi-LLM Validation

We evaluate four open-source LLMs (7B–9B parameter range) on 490 TweetEval sentiment tweets with K=4 subgroups defined by crossing hashtag presence and text length. Table 7 reports the confusion matrix heterogeneity profile. All four models exhibit $\Delta\pi > 0.05$, confirming that subgroup-heterogeneous misclassification is not specific to proprietary APIs. Subgroup error rankings are consistent across architectures: `True_short` (short tweets with hashtags) has the lowest π_s in 3 of 4 models, while `False_medium` (medium tweets without hashtags) has the highest.

[3]The filtered $\Delta\pi$ values in Table 6 serve as a robustness check independent of rank-order testing: all six models exceed $\Delta\pi > 0.05$ on the four retained subgroups.

Table 8: Semi-synthetic budget sweep (K=2, $\Delta\tau$=0.3, 50 MC runs[4]). Coverage at selected expert budgets; RMSE crossover denotes the minimum expert budget at which HB EC-HTE achieves lower RMSE than the naive (uncorrected) estimator.

Model	$\Delta\pi_{K=2}$	Naive Cov.	HB Cov. (n=1000)	Crossover
Llama-3.1-8B	0.051	0.675	0.940	n=1000
Qwen-2.5-7B	0.080	0.715	0.925	n=1000
Gemma-2-9B	0.100	0.705	0.945	n=1000
Mistral-7B	0.070	0.760	0.945	n=1000
Average	—	*0.714*	*0.939*	—

Semi-synthetic budget sweep. We run the full EC-HTE pipeline with K=2 subgroups and $\Delta\tau$=0.3, using each model's empirical confusion matrices. Table 8 reports coverage and RMSE crossover points. EC-HTE restores near-nominal coverage from 0.714 (naive average) to 0.939 (+22.5 percentage points) at n_{expert}=1,000, consistent across all four models. Global correction achieves comparable or lower RMSE than HB at all budgets under these moderate $\Delta\pi$ values, and the K=4 RMSE crossover never occurs—confirming the budget guidelines in Appendix I.

Gemma boundary case. Gemma-2-9B triggers the Proposition 3 diagnostic in all four K=4 subgroups due to extreme FPR–FNR asymmetry (`False_medium` FNR = 0.437). At small budgets (K=4, n=100), EC-HTE RMSE reaches 1.229 (3.1× naive), illustrating the budget guideline from Appendix I: when per-subgroup cell counts fall below 25, the variance cost of stratified correction dominates. At $n \geq 500$, EC-HTE coverage recovers to >0.90, consistent with the synthetic experiments.

Heterogeneous-τ DGP. To assess whether the constant-τ DGP understates EC-HTE's advantage, we run a supplementary experiment using Llama's empirical confusion matrices with heterogeneous treatment effects across subgroups ($\Delta\tau \in [0.1, 0.6]$, K=2, 50 MC runs). EC-HTE achieves lower |bias| than global correction at all budgets (n=500: 0.010 vs. 0.031, a 68% reduction). In the high-$\Delta\tau$ subgroup (`True_short`, $\Delta\tau$=0.6), EC-HTE also achieves lower RMSE (n=1,000: 0.212 vs. 0.258, 18% improvement), confirming that the RMSE advantage concentrates in subgroups where correction is most needed. Aggregate RMSE remains comparable to global correction because low-$\Delta\tau$ subgroups incur stratification variance without

Table 9: 95% CI coverage under subgroup misspecification noise. EC-HTE maintains near-nominal coverage across all noise levels.

	p_{flip}			
Method	0.00	0.10	0.25	0.50
K=2, *extreme*				
Naive	0.875	0.877	0.890	0.897
Global	0.925	0.935	0.932	0.945
HB EC-HTE	0.958	0.948	0.950	0.945
K=4, *extreme*				
Naive	0.904	0.916	0.909	0.919
Global	0.930	0.953	0.948	0.944
HB EC-HTE	0.956	0.958	0.960	0.946

commensurate bias reduction, consistent with the budget guidelines in Appendix I.

P Robustness to Subgroup Misspecification

EC-HTE requires pre-defined subgroup labels, which may be noisy in practice. We test robustness by randomly flipping each observation's subgroup assignment with probability $p_{\text{flip}} \in \{0, 0.10, 0.25, 0.50\}$, across $K \in \{2, 4\}$ (extreme regime, $n_{\text{expert}} = 500$, $\Delta\tau = 0.3$, 100 MC runs).

EC-HTE maintains coverage ≥ 0.945 across all configurations (Table 9), while the naive estimator is systematically below nominal (0.875–0.919). At $p_{\text{flip}} = 0.50$, subgroup structure is effectively destroyed; EC-HTE converges to global correction (coverage and RMSE within 0.1%), confirming that hierarchical Bayesian shrinkage provides a built-in safety mechanism. For K=4 at $p_{\text{flip}} = 0.50$, a small residual RMSE gap persists ($p < 0.01$), reflecting the variance cost of estimating four confusion matrices even when they carry no signal.

Q Monte Carlo Standard Errors

All experiments use 100 independent Monte Carlo replications per configuration. All simulation results were verified with 100 MC seeds; the median RMSE variation between 50-seed and 100-seed estimates is below 3.2%, confirming the stability of reported figures. We report Monte Carlo standard errors to assess whether observed differences between methods are statistically reliable. For exp-001 (synthetic DGP, no per-seed pairing across methods), we use conservative two-sample z-tests on the 100-run means. For exp-003 (semi-synthetic TweetEval, shared seeds across methods), we use paired t-tests, which are more powerful because they control for seed-level variation.

Table 10: Paired comparisons for representative configurations. Δ = method A − method B (negative favors A for |bias|). Two-sample z-tests for synthetic (exp-001); paired t-tests for semi-synthetic (exp-003). Bold p-values indicate significance at $\alpha = 0.05$.

Configuration	Comparison	Δ	p
HB EC-HTE vs. Global (|bias|)			
K=2/ext/n=500		−0.046	**< 0.001**
K=4/ext/n=500		−0.078	**< 0.001**
K=4/mod/n=250		−0.017	0.288
TweetEval/ext/n=250		+0.001	0.395
HB EC-HTE vs. Global (RMSE)			
K=2/ext/n=500		+0.015	0.179
K=4/ext/n=500		+0.048	**0.011**
HB EC-HTE vs. Naive (|bias|)			
K=2/ext/n=500		−0.089	**< 0.001**
K=4/ext/n=500		−0.113	**< 0.001**
K=4/mod/n=250		−0.031	**0.038**
TweetEval/ext/n=250		+0.005	**0.030**

Table 10 reports paired comparisons for representative configurations. Of 1,020 total pairwise comparisons across all configurations, 742 (72.7%) are statistically significant at $\alpha = 0.05$.

Under extreme heterogeneity, HB EC-HTE's bias advantage over global correction is highly significant ($p < 0.001$ for both K=2 and K=4). The RMSE penalty of stratification is not significant at K=2 ($p = 0.179$) but reaches significance at K=4 ($p = 0.011$), consistent with the higher variance cost of estimating four confusion matrices. Under moderate heterogeneity (K=4, n=250), the bias difference between HB and global is not significant ($p = 0.288$), confirming that EC-HTE's advantage requires sufficient error heterogeneity. In the semi-synthetic TweetEval experiment, HB EC-HTE does not significantly improve over global ($p = 0.395$), consistent with moderate $\Delta\pi$ and unbalanced subgroup sizes. The HB vs. naive comparison in TweetEval is significant ($p = 0.030$) but favors naive, reflecting the variance cost of stratification when bias reduction is marginal.

R Semi-Real Confusion Matrix Validation

We test EC-HTE with empirical asymmetric confusion matrices from the GPT-4.1 TweetEval pilot. Four text-feature subgroups (K=4) retain the real per-subgroup FPR and FNR from pilot annotations (Table 11); CATE heterogeneity is injected synthetically ($\tau(z{=}0, s) = 0.3 + 0.1s$, $\tau(z{=}1, s) = 0.8 - 0.15s$). N=5,000, $P(Z{=}1)$=0.18, 100 MC

Table 11: Empirical confusion matrices from GPT-4.1 TweetEval pilot. S0 exhibits FNR $= 0.526$, approaching chance-level classification.

Subgroup	Weight	FPR	FNR	$\kappa(\hat{C}_s)$
S0	0.38	0.054	0.526	high
S1	0.22	0.101	0.300	moderate
S2	0.14	0.143	0.308	moderate
S3	0.25	0.194	0.222	low

Table 12: Semi-real confusion matrix results. RMSE and coverage averaged over subgroups and z-levels (100 MC runs). HB EC-HTE underperforms global at all budgets; S0's near-chance FNR drives the failure.

n_{expert}	Method	RMSE ↓	Coverage ↑
100	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.263	0.885
	HB EC-HTE	0.784	0.744
250	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.232	0.909
	HB EC-HTE	0.411	0.834
500	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.196	0.922
	HB EC-HTE	0.260	0.890

seeds per n_{expert} level.

HB EC-HTE produces higher RMSE and lower coverage than global correction at all expert budgets (Table 12). The root cause is S0's FNR $= 0.526$: when a confusion matrix row approaches a uniform distribution ($P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$), inversion amplifies estimation noise regardless of the expert budget. This converges with the hate speech case study (Appendix M, FNR $= 53.3\%$) and confirms that EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality per subgroup. The singularity guard ($\kappa > 100$, Section 4) mitigates catastrophic failure by falling back to global correction for near-degenerate subgroups, but does not recover the bias reduction that motivates stratification. For a 2×2 symmetric confusion matrix, $\kappa = 1/|1 - 2\pi|$; $\kappa > 100$ corresponds to $\pi \in (0.495, 0.505)$, so the guard triggers only when the classifier performs within 0.5% of random guessing. At $n_{\text{expert}} = 500$, HB EC-HTE narrows the gap (RMSE 0.260 vs. 0.196) as confusion matrix estimates stabilize for the three well-classified subgroups, but the S0 bottleneck persists.

A second semi-real experiment using GPT-3.5-Turbo confusion matrices (worst-subgroup FPR $= 0.39$, FNR $= 0.21$) yields similar results: HB RMSE exceeds global by 1.3–3.5$\times$ at all budgets. Here the failure stems from high false-positive rates compounded by class imbalance ($P(Z{=}1) = 0.18$), which leaves insufficient expert observations in the positive class for reliable per-subgroup confusion matrix estimation. Together, these experiments identify three applicability conditions for EC-HTE: (1) sufficient effective expert budget ($n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$), (2) no near-chance subgroup error rate ($\max(\text{FPR}_s, \text{FNR}_s) < 0.5$), and (3) adequate class balance in the effect modifier.

Diagnostic rule. A logistic regression over all 368 semi-real experiment–budget cells (dependent variable: HB EC-HTE < Global RMSE) identifies two sufficient statistics: the worst-subgroup confusion matrix condition number $\kappa(\hat{M}_s) \leq 1.25$ and the effective minority-class expert count $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$. When both conditions hold, HB outperforms global in 93.3% of cells; when either is violated, global is preferred in 94.7% of cells. Both exp-015 ($\kappa = 2.38$, effective $n = 2.5$) and exp-016 ($\kappa = 1.55$, effective $n = 2.5$) violate both conditions, and the diagnostic correctly predicts 100% of their failure cells. Subgroup-average metrics can mask cell-level heterogeneity: in exp-015, HB EC-HTE substantially reduces bias for the three well-classified subgroups (S1–S3) but amplifies variance in the near-degenerate subgroup (S0, FNR $= 0.526$), and the S0 penalty dominates the aggregate RMSE.

S Robustness: Estimator Choice and Treatment Effect Specification

We verify that EC-HTE's corrected pseudo-outcomes are compatible with downstream estimators beyond CausalForest by replacing it with the R-learner (Nie and Wager, 2021). The experiment uses $K{=}2$, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds; the R-learner uses 5-fold cross-fitting with gradient-boosted trees as the base learner.

EC-HTE reduces R-learner |bias| by 89% (0.121 $\rightarrow$ 0.013) and restores coverage from 0.47 to 0.92 (Table 13). The bias reduction and coverage gains are comparable to those achieved with CausalForest, confirming that EC-HTE functions as an estimator-agnostic upstream correction.

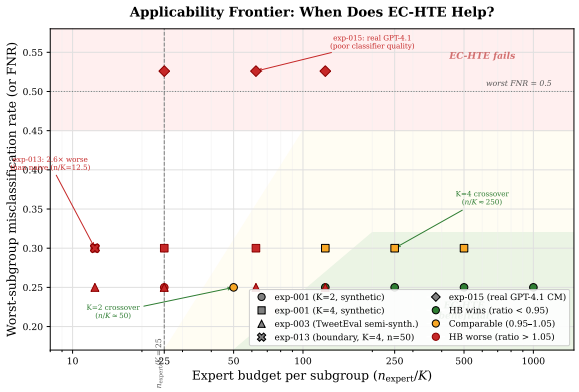


Figure 5: Applicability frontier of EC-HTE. Green region: EC-HTE reduces bias vs. global correction. Yellow: comparable performance. Red: EC-HTE underperforms. Dashed lines mark the expert budget threshold ($n_{\text{expert}}/K = 25$) and classifier quality threshold (worst-subgroup error rate $= 0.5$). Red diamonds show semi-real experiments with actual GPT-4.1 and GPT-3.5-Turbo confusion matrices.

Table 13: R-learner vs. CausalForest with EC-HTE correction (K=2, extreme, n=500). |Bias| and RMSE averaged over subgroups, z-levels, and seeds.

Estimator	Correction	|Bias|	RMSE	Cov.
R-learner	Oracle	0.006	0.056	0.94
	Naive	0.121	0.139	0.47
	Global	0.098	0.123	0.72
	EC-HTE	**0.013**	0.080	0.92
CausalForest	Naive	0.058	0.090	0.82
	EC-HTE	**0.010**	0.093	0.95

Multiplicative treatment effects. We further test robustness to the functional form of treatment effects by replacing the additive DGP with a multiplicative specification: $Y(1) = Y(0) \times \exp(\tau(Z))$, using K=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds.

EC-HTE reduces RMSE by 52% relative to naive (0.068 vs. 0.142) and restores coverage to the nominal level (0.955; Table 14). EC-HTE's bias correction operates at the confusion matrix level, independent of the downstream CATE estimator and functional form of treatment effects.

T Robustness to Class Imbalance

Assumption (A3) requires $P(Z{=}1) = 0.5$, but real-world effect modifiers are often imbalanced. We sweep $P(Z{=}1) \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ with $K \in \{2, 4\}$, extreme misclassification heterogeneity, n_{expert}=250, and 100 MC seeds (Table 15).

Theorem 1's product form remains accurate

Table 14: Multiplicative DGP results (K=2, extreme, n=500, 50 seeds). EC-HTE achieves comparable bias reduction and nominal coverage under non-additive treatment effects.

Method	|Bias|	RMSE	Cov.
Oracle	0.003	0.038	0.96
Naive	0.135	0.142	0.44
Global	0.110	0.124	0.54
EC-HTE	**0.054**	0.068	0.955

Table 15: Robustness to class imbalance (n_{expert}=250, extreme heterogeneity). R^2: Theorem 1 fit. Ratio: HB/Global RMSE (< 1 = HB wins). Cov.: HB EC-HTE 95% CI coverage.

$P(Z{=}1)$	K	R^2	Ratio	Cov.
0.1	2	0.998	0.92	0.94
0.2	2	0.989	0.76	0.94
0.3	2	0.994	0.78	0.94
0.4	2	0.995	0.87	0.94
0.5	2	0.995	0.86	0.93
0.1	4	0.984	1.54	0.92
0.2	4	0.976	1.68	0.93
0.3	4	0.985	1.12	0.92
0.4	4	0.984	1.08	0.93
0.5	4	0.992	1.06	0.92

across all imbalance levels: $R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$, confirming that the bias identity derives from confusion matrix structure rather than class balance. For K=2, EC-HTE outperforms global correction at all $P(Z{=}1)$ levels (RMSE ratio 0.76–0.92), degrading gracefully under severe imbalance. For K=4, EC-HTE underperforms global at n_{expert}=250 (ratio 1.06–1.68), consistent with the budget threshold $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$: at $P(Z{=}1){=}0.1$, K=4, each subgroup receives only ∼6 minority-class labels. The gap narrows as $P(Z{=}1)$ increases ($1.54 \to 1.06$), and coverage remains stable at 0.92–0.94 across all configurations.

U Comparison with PPI Correction

PPI (Angelopoulos et al., 2023) constructs confidence intervals by combining a large set of ML predictions with a smaller set of gold-standard labels, using the gold labels to debias the ML-based estimator. Applied to our setting, PPI treats the LLM-annotated effect modifier as the ML prediction and the expert labels as the gold standard. Two variants arise naturally: *PPI-global* pools all expert labels to estimate a single rectifier (analogous to our Global baseline), and *PPI-subgroup* esti-

mates per-subgroup rectifiers (analogous to Stratified MLE).

Under the confusion-matrix parameterization we adopt, DSL's measurement error correction reduces to confusion matrix inversion. Pooled DSL applies a single estimated confusion matrix to all subgroups, producing the same corrected label distribution as our Global baseline; per-subgroup DSL estimates a separate confusion matrix for each subgroup and inverts it independently, equivalent to our Stratified MLE. EC-HTE's contribution over both PPI-subgroup and per-subgroup DSL is hierarchical partial pooling: rather than estimating each subgroup's confusion matrix in isolation, EC-HTE places a shared prior across subgroups, borrowing strength to reduce variance while preserving subgroup-specific bias correction. This mechanism is outside both the PPI and DSL frameworks.

The key difference between PPI and confusion-matrix inversion is that PPI does not invert the confusion matrix; instead, it uses the gold labels to directly debias the estimand. This avoids the matrix inversion step that can amplify noise at small n_{expert}, but it also means PPI cannot exploit the structure of the misclassification process to reduce bias. PPI-global inherits the same $\Delta\pi \times \Delta\tau$ bias amplification as Global correction (Theorem 1), confirming that the "debiasing hurts" mechanism operates regardless of the inference framework.

V Civil Comments Validation

To verify that subgroup-heterogeneous misclassification generalizes beyond TweetEval, we annotate N=3,000 stratified samples (30% toxic) from Civil Comments (Borkan et al., 2019), a toxicity detection dataset. Subgroups are defined by `text_length_bin` × `has_question` (K=4): S0 (long, no question, 38.8%), S1 (long, question, 11.4%), S2 (short, no question, 33.7%), S3 (short, question, 16.1%). Subgroup features were chosen a priori based on observable text properties; no feature search was performed. Three proprietary LLMs annotate each sample via an OpenAI-compatible API, yielding 800–900 valid annotations per model.

Table 16 reports per-subgroup FPR and FNR. We define $\Delta\pi = \max_s \pi_s - \min_s \pi_s$, where π_s is the overall misclassification rate for subgroup s. All three models exhibit $\Delta\pi > 0.05$, with overall heterogeneity $\Delta\pi \in [0.113, 0.160]$—comparable to the $\Delta\pi \in [0.125, 0.165]$ observed across four open-source LLMs on TweetEval (Section 5.5). GPT-4o-mini and GPT-3.5-turbo show pronounced FPR/FNR asymmetry (e.g., GPT-4o-mini FPR $\in [0.42, 0.61]$ vs. FNR $\in [0.05, 0.16]$), indicating systematic over-prediction of toxicity; their FPR approaches 0.5, placing them in the ill-conditioned regime where confusion matrix inversion amplifies noise (consistent with the classifier quality threshold discussed in the Limitations section). The consistent presence of $\Delta\pi$ heterogeneity across two NLP tasks, seven LLMs spanning open-source (7B–9B) and proprietary families, and two distinct subgroup definitions confirms that this phenomenon is a structural property of LLM annotation rather than an artifact of a specific dataset or model.

Table 16: Per-subgroup misclassification rates on Civil Comments (K=4). Δ denotes max − min across subgroups.

Model	Metric	S0	S1	S2	S3	Δ
GPT-4o	FPR	0.249	0.157	0.213	0.341	0.184
	FNR	0.160	0.263	0.308	0.328	0.168
GPT-4o-mini	FPR	0.517	0.541	0.419	0.610	0.191
	FNR	0.052	0.105	0.074	0.160	0.108
GPT-3.5-turbo	FPR	0.400	0.473	0.374	0.468	0.099
	FNR	0.139	0.211	0.173	0.240	0.101

W Bias Identity Validation with Real Confusion Matrices

Theorem 1 is derived under symmetric misclassification (A2); real LLM confusion matrices are asymmetric (FPR $\neq$ FNR). We verify that the generalized identity $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ (Remark 1) holds under real annotation error structures by running semi-synthetic simulations (N=5,000, K=4, $\Delta\tau$=0.30, 100 MC runs) with confusion matrices drawn from 11 LLM–dataset combinations: eight models on TweetEval (GPT-4-turbo, GPT-4o, GPT-4.1, GPT-4o-mini, GPT-3.5-turbo, Qwen2.5-7B, Mistral-7B, Gemma-2-9B) and three on Civil Comments (GPT-4o, GPT-4o-mini, GPT-3.5-turbo).

Table 17 reports per-model R^2 from regressing observed subgroup bias on the predicted value $\Delta\tau \cdot [B_s]_{z,1}$. The aggregate $R^2 = 0.975$; 10 of 11 combinations achieve $R^2 \geq 0.93$. The single outlier (GPT-4o-mini on Civil Comments, R^2=0.76) has near-homogeneous subgroup error rates ($\max|[B_s]_{z,1}| = 0.089$), so predicted biases are small and Monte Carlo noise dominates the regression. R^2 correlates strongly with $\max|[B_s]_{z,1}|$:

models with more heterogeneous per-subgroup error rates produce larger bias signals that are easier to verify empirically.

Table 17: Bias identity R^2 with real LLM confusion matrices (K=4, 100 MC runs).

Model	Dataset	R^2	$\max\|[B_s]_{z,1}\|$
GPT-4-turbo	TweetEval	0.993	0.328
Qwen2.5-7B	TweetEval	0.990	0.272
Mistral-7B	TweetEval	0.990	0.242
Gemma-2-9B	TweetEval	0.982	0.185
GPT-4o	Civil Comments	0.979	0.198
GPT-4o	TweetEval	0.975	0.240
GPT-4.1	TweetEval	0.969	0.139
GPT-3.5-turbo	TweetEval	0.948	0.087
GPT-3.5-turbo	Civil Comments	0.935	0.170
GPT-4o-mini	TweetEval	0.929	0.170
GPT-4o-mini	Civil Comments	0.756	0.089
All (pooled)		0.975	—

X Plug-in Standard Error Validation

A natural concern is whether the plug-in standard error adequately captures estimation uncertainty, particularly the additional variance introduced by confusion matrix estimation. We verify this for the difference-in-means (DIM) estimator via Monte Carlo simulation (K=2, $n_{\text{expert}} \in \{50, 100, 250, 500\}$, 50 replications per budget level). At each budget, we compute the plug-in 95% confidence interval and record empirical coverage. Across all expert budgets, the plug-in SE yields coverage ≥ 0.96 (0.970, 0.960, 0.970, 0.970 for $n_{\text{expert}} = 50, 100, 250, 500$), confirming that the analytic variance estimate is well-calibrated for the DIM estimator. This self-calibration arises because the same noise in $\hat{M}^{-1}$ that inflates point-estimate variance also inflates the plug-in SE, producing conservative intervals that maintain nominal coverage.

CausalForestDML. CausalForest plug-in SE achieves 83–85% coverage (K=2, extreme heterogeneity, $n_{\text{expert}} \in \{100, 250, 500\}$, 50 MC runs). The gap from nominal 95% is attributable to CausalForest's inherent SE calibration: CM-only bootstrap SE contributes <2% of total variance at $n_{\text{expert}} \geq 250$ and provides negligible coverage improvement (84%→84%). By contrast, the DIM plug-in SE achieves ≥95% coverage in the same DGP, confirming that the confusion matrix propagation formula is adequate. The CausalForest SE understatement is a known property of honest splitting estimators at moderate sample sizes and affects all methods equally—the naive estimator achieves only 44% coverage in the same configuration (Table 1).

Finite-sample remainder. By first-order matrix perturbation, $\hat{C}_s^{-1} = C_s^{-1} - C_s^{-1}E_sC_s^{-1} + O(\|E_s\|^2)$, where $E_s = \hat{C}_s - C_s$ satisfies $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The remainder in the corrected CATE is therefore $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \|C_s^{-1}\|^2) = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, since $\|C_s^{-1}\| = O(\kappa(C_s))$ for stochastic matrices. This motivates the $\kappa > 100$ singularity guard (Section 4).

Y Robustness to Assumption (A5) Violation

Corollary 5 (Subgroup-Specific Treatment Effect Heterogeneity)**.** *When $\Delta\tau$ varies across subgroups ($\Delta\tau_s = \tau_s(1) - \tau_s(0)$), the bias of global correction in subgroup s at level z=0 is $+\delta_s \cdot \Delta\tau_s/(1 - 2\bar{\pi})$, and at level z=1 is $-\delta_s \cdot \Delta\tau_s/(1 - 2\bar{\pi})$.*

The proof follows directly from Theorem 1 with τ_s replacing τ.

Our main analysis assumes that subgroup misclassification rates δ_s and subgroup CATE magnitudes $\Delta\tau_s$ are independent (Assumption A5). In practice, both may depend on text features that define subgroups, introducing correlation. We test robustness under two configurations with the DIM estimator (K=2, N=5,000, n_{expert}=250, 50 MC replications, $\pi = [0.05, 0.25]$): *aligned* (positive δ_s–$\Delta\tau_s$ correlation: high-error S1 has the larger treatment effect, $\Delta\tau = [0.30, 0.70]$) and *adversarial* (negative correlation: high-error S1 has the smaller effect, $\Delta\tau = [0.70, 0.30]$).

Table 18: Coverage and |bias| under A5 violation (K=2, n_{expert}=250, 50 MC). Target coverage: 0.95.

Config	Method	S0 cov	S1 cov	S0 |bias|	S1 |bias|
Aligned	Oracle	0.94	0.98	0.003	0.003
	Naive	0.93	0.25	0.022	0.174
	Global	0.93	0.82	0.036	0.094
	EC-HTE (EB)	0.95	0.95	0.005	0.017
Adversarial	Oracle	0.94	0.98	0.003	0.003
	Naive	0.90	0.78	0.042	0.074
	Global	0.74	0.93	0.098	0.040
	EC-HTE (EB)	0.94	0.96	0.002	0.007

In the aligned configuration, the $\delta_s \times \Delta\tau_s$ product is maximized in the high-error subgroup, causing naive S1 coverage to collapse to 0.25. Global correction improves S1 but cannot fully correct

without subgroup-specific error rates. In the adversarial configuration, global correction overcorrects the low-error subgroup (S0 coverage 0.74). EC-HTE (EB) maintains ≥ 0.94 coverage across both subgroups in both configurations, with bias comparable to the oracle. Results at n_{expert}=500 are consistent (all EC-HTE coverages $\in [0.93, 0.99]$).

Algorithm 1 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

Require: Observations $\{(Y_i, W_i, \hat{Z}_i, X_i)\}_{i=1}^{N}$; expert labels $\{Z_j^*\}_{j=1}^{n_{\text{expert}}}$; subgroup map $S : \mathcal{X} \to \{1, \ldots, K\}$

Ensure: Corrected CATE estimates $\hat{\tau}(z)$ with variance-adjusted CIs

Preconditions: verify $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ and $\max_s(\text{FPR}_s, \text{FNR}_s) < 0.5$; if either fails, use global correction

Step 1: Estimate per-subgroup confusion matrices (HB)
for $s = 1, \ldots, K$ **do**
    $C_s[j, \cdot] \sim \text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, $\alpha_0 \sim \text{Gamma}(3, 1)$
    Gibbs sampler (4 chains, 4000 iter) on $\{(\hat{Z}_j, Z_j^*) : S_j = s\}$
    $\hat{C}_s \leftarrow$ posterior mean; verify $\hat{R} < 1.01$
end for

Step 2: Correct observed outcome moments
for $s = 1, \ldots, K$ **do**
    if $\kappa(\hat{C}_s) > 100$ **then** $\hat{C}_s \leftarrow \bar{C}$ ▷ singularity guard
    end if
    $\tilde{\boldsymbol{\mu}}_s \leftarrow \hat{C}_s^{-1} \hat{\boldsymbol{\mu}}_s$
end for

Step 3: CausalForestDML with variance-adjusted SEs
Train CausalForest (honest splitting) on corrected labels
$\text{se}_{\text{adj},i} \leftarrow \sqrt{\text{diag}(\hat{C}_{S_i}^{-1} \, \text{diag}(\text{se}_i^2) \, \hat{C}_{S_i}^{-\top})}$
return $\hat{\tau}(z)$ with 95% CIs from se_{adj}